
name: aesc-terminal-orchestrator description: Operational guide and prompt skill for LLMs to interface with the Æsc Terminal Daemon APK, routing NPU models via UNIX domain sockets, executing shell commands through local ADB WebSockets, and retrieving files via MCP.

Æsc Terminal Orchestrator Protocol

Overview & Identity

The **Æsc Terminal Daemon** (`aesc` APK) is the native Android system execution layer that completely replaces Termux. It runs as a `START_STICKY` foreground Android service, holds the local ADB loopback connection on `127.0.0.1:5555` with elevated `UID 2000 (shell)` privileges, hosts a Ktor WebSocket server on port `8080`, and interfaces directly with the Qualcomm Hexagon HTP v79 NPU over UNIX domain sockets.

When operating as an agent inside or connected to the Æsc terminal environment, you must adhere to the three operational execution channels detailed below.

Execution Channel 1: Snapdragon Hexagon NPU Routing

All local model swaps and context priority allocations are routed through the local UNIX domain socket:

- **Socket Path:** `/dev/socket/npu_manager.sock`
- **Transport:** POSIX AF_UNIX Stream Socket

1. Model Roles & Hardware Priorities

- **0.8B_triage** (`qwen3.5-0.8b-q4_0`): Run under `QNN_PRIORITY_LOW`. Used for intent routing, token classification, and entity extraction. Interruptible with minimal thermal impact.
- **9B_executor** (`qwen3.5-9b-q4_0`): Run under `QNN_PRIORITY_HIGH`. Used for code generation, multi-step planning, and complex AST analysis.

2. Hardware Prefill Chunking Guardrail

- **Maximum Prefill Chunk:** 512 tokens.
- Never send unchunked prompts larger than 512 tokens directly to the HTP in a single pass. Hexagon DSP hardware watchdog timers will abort execution if prefill bursts exceed hardware timeout thresholds.

3. Socket JSON-RPC Schema

To dispatch an inference pass or check status, format your socket payload as:

```
{  
  
  "action": "infer",  
  
  "model_role": "0.8B_triage",  
  
  "tokens": [101, 2054, 2003, ...],  
  
  "max_tokens": 1024  
}
```

To request a model swap:

```
{  
  
  "action": "swap_model",  
  
  "target_role": "9B_executor"  
}
```

Execution Channel 2: Local ADB Loopback Shell

Terminal shell commands must be executed through the elevated ADB loopback bridge rather than standard Android app sandboxes.

1. Connection Architecture

- **Loopback URL:** `ws://127.0.0.1:8080/shell`

- **Underlying ADB Bridge:** `127.0.0.1:5555` (Paired via Wireless Debugging)
- **Privilege Level:** `UID 2000 (shell)` — grants access to background daemon spawning, `/data/local/tmp`, and cross-process signals without requiring root.

2. Execution Protocol

1. Open a WebSocket frame to `ws://127.0.0.1:8080/shell`.
2. Transmit the command string as a plain text frame.
3. Stream the output frames until the process exit code is returned.
4. **Safety Rule:** Always use absolute paths or paths rooted at `${HOME}/novae-xorpus`. Never issue recursive deletion commands (`rm -rf`) on root or sensory archives (`01_raw_sources/`).

Execution Channel 3: Model Context Protocol (MCP) Suite

When retrieving files, inspecting schemas, or updating memory records, use the configured MCP connectors rather than raw terminal calls whenever possible.

Available MCP Servers

1. **filesystem:**
 - Scoped to `${HOME}/novae-xorpus`.
 - Tools: `read_file`, `write_file`, `list_directory`, `get_file_info`.
 - Invariant: Modification of files in `01_raw_sources/` is strictly denied.
2. **sqlite_memory:**
 - Connected to `${HOME}/novae-xorpus/_dumbass_universal_memory/sqlite/aesc_cache.db`.
 - Tools: `query`, `execute`, `get_task_state`.
 - Used for low-latency active session state persistence and entity lookups.
3. **npu_manager:**
 - UNIX socket bridge for real-time model telemetry and thermal status checks.

Summary of Invariants for Agents

1. **Thermal Headroom Check:** Before launching heavy 9B executor passes, query `/dev/socket/npu_manager.sock` for thermal state. If status is `WARM` ($\geq 42^{\circ}\text{C}$), defer non-critical batch processing.

2. **Zero App GC Interference:** Keep model execution completely inside native background processes (`:qairt_engine`, `qnn_http_scheduler.py`) to prevent Android UI micro-stutters.
3. **Session Checkpointing:** Log all multi-step tool invocations to `sqlite_memory` so that any unexpected daemon termination resumes from the latest verified step.